36. Ethernet Controller (ETHERC)

36.1 Overview

This MCU has a two-channel Ethernet controller (ETHERC) compliant with the Ethernet or IEEE802.3 Media Access Control (MAC) layer protocol. Each ETHERC channel has one channel of the MAC layer interface, and connecting the MCU to the physical layer LSI (PHY-LSI) allows transmission and reception of frames compliant with the Ethernet/ IEEE802.3 standard. The ETHERC is connected to the Ethernet controller direct memory access controller (EDMAC), so data can be transferred without using the CPU.

Table 36.1 lists the ETHERC specifications. Figure 36.1 shows the ETHERC configuration. Table 36.2 lists the ETHERC I/O pins.

Figure 36.2 and Figure 36.3 show examples of connecting the MCU to the PHY-LSI.

Table 36.1 ETHERC Specifications

Item	Description
Number of channels	Two channels
Protocol	Flow control compliant with IEEE802.3x
Data transmission/reception	Frames compliant with the Ethernet/IEEE802.3 standard can be transmitted and received.
Bit rate	Supports 10 Mbps and 100 Mbps
Operation modes	Supports full-duplex and half-duplex modes
Interfaces	Media Independent Interface (MII), Reduced Media Independent Interface (RMII), compliant with the IEEE802.3u standard
Functions	Magic Packet™ *1 detection, Wake-On-LAN (WOL) signal output

Note 1. Magic Packet is a trademark of Advanced Micro Devices, Inc.

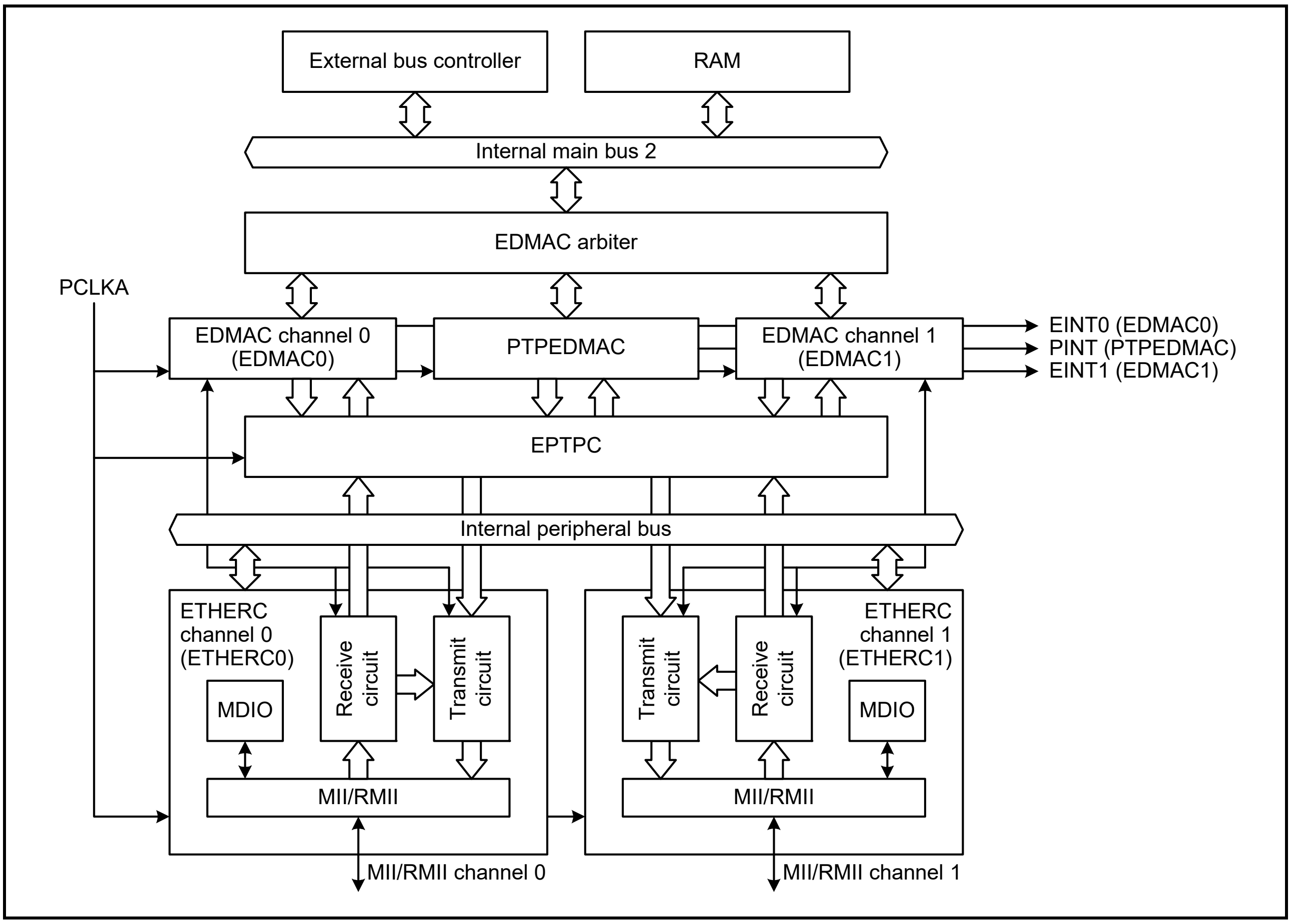


Figure 36.1 ETHERC Configuration

Table 36.2 ETHERC I/O Pins (n = 0, 1)

Operating Mode	Pin Name	I/O	Description
MII	ETn_TX_CLK *1	Input	Transmit clock Timing reference signal for outputting the ETn_TX_EN, ETn_ETXD3 to ETn_ETXD0, and ETn_TX_ER signals.
	ETn_RX_CLK *1	Input	Receive clock Timing reference signal for inputting the ETn_RX_DV, ETn_ERXD3 to ETn_ERXD0, and ETn_RX_ER signals.
	ETn_TX_EN *1	Output	Transmit data valid This signal indicates that valid transmit data has been output on pins ETn_ETXD3 to ETn_ETXD0.
	ETn_ETXD3 to ETn_ETXD0 *1	Output	4-bit transmit data
	ETn_TX_ER *1	Output	Transmit error This signal notifies the PHY-LSI that an error occurred during transmission. This signal is fixed to low in this MCU.
	ETn_RX_DV *1	Input	Receive data valid This signal indicates that valid receive data is on pins ETn_ERXD3 to ETn_ERXD0.
	ETn_ERXD3 to ETn_ERXD0 *1	Input	4-bit receive data
	ETn_RX_ER *1	Input	Receive error This signal indicates that there is an error in a frame that is being transferred from the PHY-LSI to the ETHERC.
	ETn_CRS *1	Input	Carrier sense
	ETn_COL *1	Input	Collision detection signal
	ETn_MDC *1	Output	Management data clock Reference clock signal for transfer of information on the ETn_MDIO pin.
	ETn_MDIO *1	I/O	Management data I/O Bidirectional data signal for exchanging management data with the PHY-LSI.
	ETn_LINKSTA	Input	Link status input from the PHY-LSI
	ETn_EXOUT	Output	General output pin
	ETn_WOL	Output	Wake-On-LAN. This signal indicates that a Magic Packet has been received.
RMII	REF50CKn *2	Input	Reference clock Timing reference signal for pins RMIIn_TXD_EN, RMIIn_TXD1 to RMIIn_TXD0, RMIIn_CRS_DV, RMIIn_RXD1 to RMIIn_RXD0, and RMIIn_RX_ER.
	RMIIn_TXD_EN *2	Output	Transmit data valid This signal indicates that valid transmit data has been output on pins RMIIn_TXD1 and RMIIn_TXD0.
	RMIIn_TXD1 to RMIIn_TXD0 *2	Output	2-bit transmit data
	RMIIn_CRS_DV *2	Input	Carrier sense/receive data valid This signal indicates that valid receive data is on pins RMIIn_RXD1 and RMIIn_RXD0.
	RMIIn_RXD1 to RMIIn_RXD0 *2	Input	2-bit receive data
	RMIIn_RX_ER *2	Input	Receive error This signal indicates that there is an error in a frame that is being transferred from the PHY-LSI to the ETHERC.
	ETn_MDC *2	Output	Management data clock Reference clock signal for transfer of information on the ETn_MDIO pin
	ETn_MDIO *2	I/O	Management data I/O Bidirectional data signal for exchanging management data with the PHY-LSI.
	ETn_LINKSTA	Input	Link status input from the PHY-LSI.
	ETn_EXOUT	Output	General output pin
	ETn_WOL	Output	Wake-On-LAN. This signal indicates that a Magic Packet has been received.

Note 1. MII signal compliant with IEEE802.3u
Note 2. RMII signal compliant with IEEE802.3u

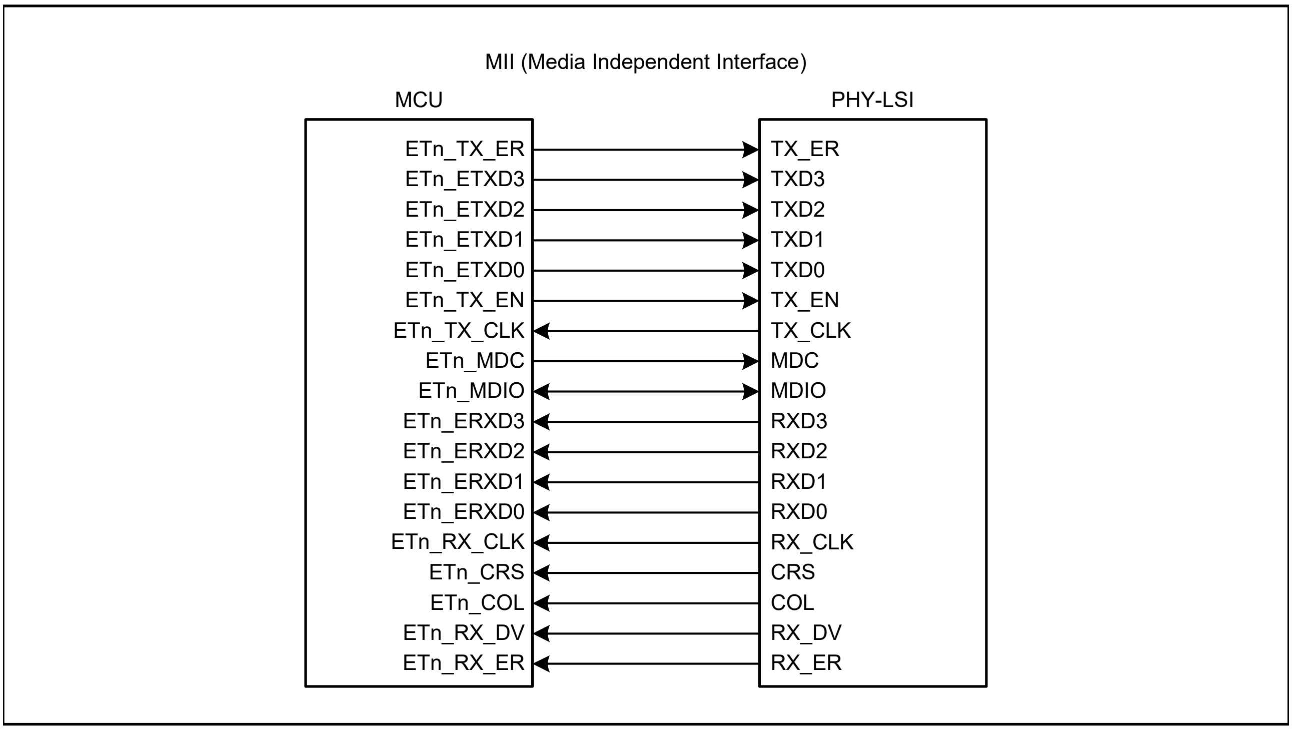


Figure 36.2 Example of Connection with the PHY-LSI for the MII

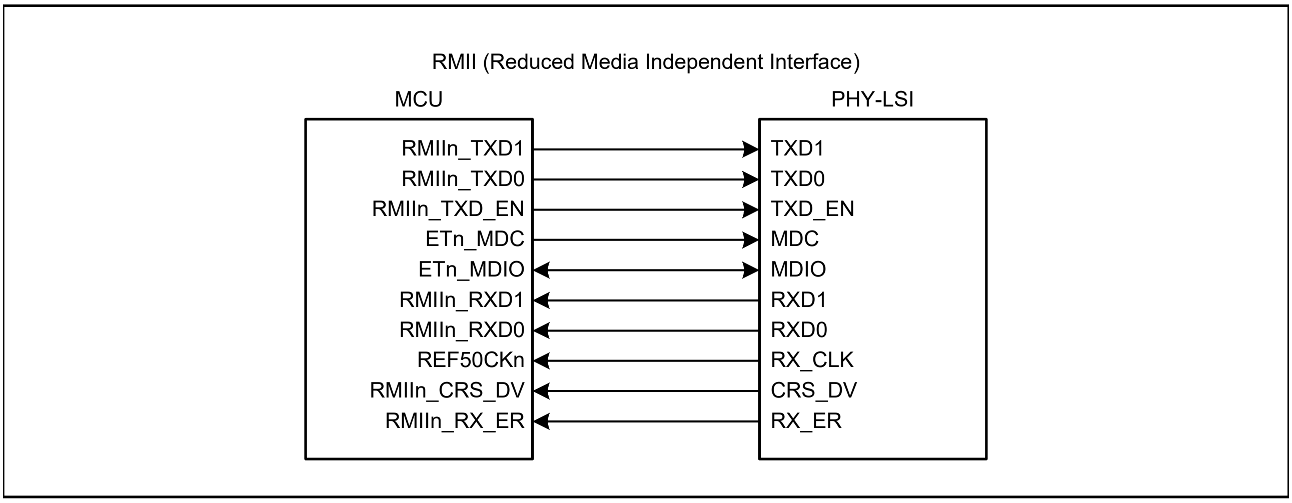


Figure 36.3 Example of Connection with the PHY-LSI for the RMII

36.2 Register Descriptions

36.2.1 ETHERC Mode Register (ECMR)

Address(es): ETHERC0.ECMR 000C 0100h, ETHERC1.ECMR 000C 0300h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	TPC	ZPF	PFR	RXF	TXF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	PRCEF	—	—	MPDE	—	—	RE	TE	—	ILB	RTM	DM	PRM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PRM	Promiscuous Mode	0: Promiscuous mode is disabled. 1: Promiscuous mode is enabled.	R/W
b1	DM	Duplex Mode	0: Half-duplex mode 1: Full-duplex mode	R/W
b2	RTM	Bit Rate	0: 10 Mbps 1: 100 Mbps	R/W
b3	ILB	Internal Loopback Mode	0: Normal data transmission or reception is performed. 1: Data is looped back in the ETHERC when full-duplex mode is selected.	R/W
b4	—	Reserved	The read value is 0. The write value should be 0.	R/W
b5	TE	Transmission Enable	0: Transmit function is disabled. 1: Transmit function is enabled.	R/W
b6	RE	Reception Enable	0: Receive function is disabled. 1: Receive function is enabled.	R/W
b8, b7	—	Reserved	The read value is 0. The write value should be 0.	R/W
b9	MPDE	Magic Packet Detection Enable	0: Magic Packet detection is disabled. 1: Magic Packet detection is enabled.	R/W
b11, b10	—	Reserved	The read value is 0. The write value should be 0.	R/W
b12	PRCEF	CRC Error Frame Receive Mode	0: EDMAC is notified of a CRC error. 1: EDMAC is not notified of a CRC error.	R/W
b15 to b13	—	Reserved	The read value is 0. The write value should be 0.	R/W
b16	TXF	Transmit Flow Control Operating Mode	0: Automatic PAUSE frame transmission is disabled. (PAUSE frame is not automatically transmitted.) 1: Automatic PAUSE frame transmission is enabled. (PAUSE frame is automatically transmitted as required.)	R/W
b17	RXF	Receive Flow Control Operating Mode	0: PAUSE frame detection is disabled. 1: PAUSE frame detection is enabled.	R/W
b18	PFR	PAUSE Frame Receive Mode	0: PAUSE frame is not transferred to the EDMAC. 1: PAUSE frame is transferred to the EDMAC.	R/W
b19	ZPF	0 Time PAUSE Frame Enable	0: PAUSE frame that contains the pause_time parameter of 0 is not used. 1: PAUSE frame that contains the pause_time parameter of 0 is used.	R/W
b20	TPC	PAUSE Frame Transmit	0: PAUSE frame is transmitted even during a PAUSE period. 1: PAUSE frame is not transmitted during a PAUSE period.	R/W
b31 to b21	—	Reserved	The read value is 0. The write value should be 0.	R/W

The ECMR register controls the ETHERC operation.

Set bits in the ECMR register, excluding bits TE and RE, during initialization after a reset. When rewriting this register outside the initialization process, set the EDMACn.EDMR.SWR bit to 1 to reset the EDMAC and ETHERC, and then set this register again.

PRM Bit (Promiscuous Mode)

When the PRM bit is set to 1, the ETHERC operates in promiscuous mode where all Ethernet frames are received. In promiscuous mode, the ETHERC receives all valid frames regardless of whether or not the address matches the destination address or broadcast address and regardless of the multicast bit setting.
When the function of the EPTPC is used (receiving PTP messages), set this bit to 1.

RTM Bit (Bit Rate)

The RTM bit sets the bit rate when the RMII is selected.

ILB Bit (Internal Loopback Mode)

When the ILB bit is set to 1, transmit frames can be looped back in the MCU. Set the DM bit to 1 (full-duplex mode) to perform a loopback test.

TE Bit (Transmission Enable)

When the TE bit is set to 1, the ETHERC transmit function is enabled.
When the TE bit is set to 0, the transmit function is disabled after the frame being processed is completely transmitted.

RE Bit (Reception Enable)

When the RE bit is set to 1, the ETHERC receive function is enabled.
When the RE bit is set to 0, the receive function is disabled after the frame being processed is completely received.

PRCEF Bit (CRC Error Frame Receive Mode)

When the PRCEF bit is set to 1, the EDMAC is not notified that a CRC error has occurred even when the error is detected in a receive frame. Accordingly, the EDMACn.EESR.CERF flag and RFS0 bit in receive descriptor 0 (RD0) do not become 1.

ZPF Bit (0 Time PAUSE Frame Enable)

When the ZPF bit is 1, a PAUSE frame that contains the pause_time parameter of 0 is transmitted when the PAUSE frame transmit request is canceled before the PAUSE time of the previously transmitted PAUSE frame has not elapsed. After the PAUSE frame that contains the pause_time parameter of 0 is received, the ETHERC is ready for transmission.
When the ZPF bit is 0, even if the PAUSE frame transmit request from the receive FIFO is canceled, the next PAUSE frame is not transmitted until the PAUSE time of the previously transmitted PAUSE frame has elapsed. When a PAUSE frame that contains the pause_time parameter of 0 is received, the PAUSE frame is discarded.

36.2.2 Receive Frame Maximum Length Register (RFLR)

Address(es): ETHERC0.RFLR 000C 0108h, ETHERC1.RFLR 000C 0308h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	RFL[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	RFL[11:0]	Receive Frame Maximum Length	The set value becomes the maximum frame length. The minimum value that can be set is 1,518 bytes, and the maximum value that can be set is 2,048 bytes. Values that are less than 1,518 bytes are regarded as 1,518 bytes, and values larger than 2,048 bytes are regarded as 2,048 bytes.	R/W
b31 to b12	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RFLR register sets the maximum frame length that can be received by the MCU. Set the length in bytes.
Do not rewrite this register while the ECMR.RE bit is 1 (receive function is enabled).

RFL[11:0] Bits (Receive Frame Maximum Length)

The RFL[11:0] bits set a frame length to be checked. When the number of bytes for fields from the destination address to the frame check sequence (FCS) of the received frame exceeds the RFL[11:0] bit value, the EDMAC is notified of a frame-too-long error.
When the received frame length exceeds the RFL[11:0] bit value, the excess data is discarded.

36.2.3 ETHERC Status Register (ECSR)

Address(es): ETHERC0.ECSR 000C 0110h, ETHERC1.ECSR 000C 0310h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	BFR	PSRTO	—	LCHNG	MPD	ICD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ICD	False Carrier Detect Flag	0: PHY-LSI has not detected a false carrier on the line. 1: PHY-LSI has detected a false carrier on the line.	R/W *1
b1	MPD	Magic Packet Detect Flag	0: Magic Packet has not been detected. 1: Magic Packet has been detected.	R/W *1
b2	LCHNG	Link Signal Change Flag	0: Change in the ETn_LINKSTA signal has not been detected. 1: Change in the ETn_LINKSTA signal has been detected (high to low, or low to high).	R/W *1
b3	—	Reserved	The read value is 0. The write value should be 0.	R/W
b4	PSRTO	PAUSE Frame Retransmit Over Flag	0: PAUSE frame retransmit count has not reached the upper limit. 1: PAUSE frame retransmit count has reached the upper limit.	R/W *1
b5	BFR	Continuous Broadcast Frame Reception Flag	0: The number of continuously received broadcast frames has not exceeded the value set in the BCFRR register. 1: The number of continuously received broadcast frames has exceeded the value set in the BCFRR register.	R/W *1
b31 to b6	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. Write 1 to clear the flag.

The ECSR register indicates the status of the ETHERC.
When any flag in the ECSR register becomes 1 while the corresponding bit in the ECSIPR register is 1 (interrupt is notified), the EDMACn.EESR.ECI flag becomes 1.

ICD Flag (False Carrier Detect Flag)

The ICD flag indicates that the PHY-LSI has detected a false carrier on the line.
The ICD flag becomes 1 when a receive error signal shown in Figure 36.10 is received from the PHY-LSI. Note that the information may not be correct when signals input from the PHY-LSI change faster than software recognizes the change. Check the timing of the PHY-LSI.

LCHNG Flag (Link Signal Change Flag)

The LCHNG flag indicates that ETn_LINKSTA signal input from the PHY-LSI has changed from high to low, or from low to high.
Refer to the PSR.LMON flag for the current link status.

PSRTO Flag (PAUSE Frame Retransmit Over Flag)

The PSRTO flag indicates that the number of retransmissions reaches the value set in the TPAUSER register when retransmitting a PAUSE frame while automatic PAUSE frame transmission is enabled.

36.2.4 ETHERC Interrupt Enable Register (ECSIPR)

Address(es): ETHERC0.ECSIPR 000C 0118h, ETHERC1.ECSIPR 000C 0318h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	BFSIPR	PSRTOIP	—	LCHNGIP	MPDIP	ICDIP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ICDIP	False Carrier Detect Interrupt Enable	0: Notification of the false carrier detect interrupt is disabled. 1: Notification of the false carrier detect interrupt is enabled.	R/W
b1	MPDIP	Magic Packet Detect Interrupt Enable	0: Notification of the Magic Packet detect interrupt is disabled. 1: Notification of the Magic Packet detect interrupt is enabled.	R/W
b2	LCHNGIP	LINK Signal Change Interrupt Enable	0: Notification of ETn_LINKSTA signal change interrupt is disabled. 1: Notification of ETn_LINKSTA signal change interrupt is enabled.	R/W
b3	—	Reserved	The read value is 0. The write value should be 0.	R/W
b4	PSRTOIP	PAUSE Frame Retransmit Over Interrupt Enable	0: Notification of PAUSE frame retransmit over interrupt is disabled. 1: Notification of PAUSE frame retransmit over interrupt is enabled.	R/W
b5	BFSIPR	Continuous Broadcast Frame Reception Interrupt Enable	0: Notification of continuous broadcast frame reception interrupt is disabled. 1: Notification of continuous broadcast frame reception interrupt is enabled.	R/W
b31 to b6	—	Reserved	The read value is 0. The write value should be 0.	R/W

The ECSIPR register selects whether or not to notify the EDMAC of the status indicated by the ECSR register. Each bit corresponds to the flag in the ECSR register that has the same bit number.

36.2.5 PHY Interface Register (PIR)

Address(es): ETHERC0.PIR 000C 0120h, ETHERC1.PIR 000C 0320h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	MDI	MDO	MMD	MDC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	x	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MDC	MII/RMII Management Data Clock	The MDC bit value is output from the ETn_MDC pin to supply the management data clock to the MII or RMII.	R/W
b1	MMD	MII/RMII Management Mode	0: Read 1: Write	R/W
b2	MDO	MII/RMII Management Data-Out	The MDO bit value is output from the ETn_MDIO pin when the MMD bit is 1 (write). The value is not output when the MMD bit is 0 (read).	R/W
b3	MDI	MII/RMII Management Data-In	This bit indicates the level of the ETn_MDIO pin. The write value should be 0.	R/W
b31 to b4	—	Reserved	The read value is 0. The write value should be 0.	R/W

The PIR register is used to access registers in the PHY-LSI via the MII or RMII. The management clock and management data are controlled by software.

Refer to section 36.3.4, Accessing MII/RMII Registers for details on accessing MII and RMII registers.

36.2.6 PHY Status Register (PSR)

Address(es): ETHERC0.PSR 000C 0128h, ETHERC1.PSR 000C 0328h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LMON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x

Bit	Symbol	Bit Name	Description	R/W
b0	LMON	ETn_LINKSTA Pin Status Flag	The link status can be read by connecting the link signal output from the PHY-LSI to the ETn_LINKSTA pin. For details on the polarity, refer to the specifications of the connected PHY-LSI.	R
b31 to b1	—	Reserved	The read value is 0. The write value should be 0.	R

The PSR register is used to monitor interface signals from the PHY-LSI.

36.2.7 Random Number Generation Counter Limit Setting Register (RDMLR)

Address(es): ETHERC0.RDMLR 000C 0140h, ETHERC1.RDMLR 000C 0340h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	RMD[19:16]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	RMD[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b19 to b0	RMD[19:0]	Random Number Generation Counter	00000h: Normal operation 00001h to FFFFFh: Setting prohibited	R/W
b31 to b20	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RDMLR register sets the maximum value for the counter used in the random number generator.
Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled) or while the ECMR.RE bit is 1 (receive function is enabled).

36.2.8 Interpacket Gap Register (IPGR)

Address(es): ETHERC0.IPGR 000C 0150h, ETHERC1.IPGR 000C 0350h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	IPG[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	IPG[4:0]	Interpacket Gap	00h: 16 bit time 01h: 20 bit time : : 14h: 96 bit time (initial value) : : 1Fh: 140 bit time	R/W
b31 to b5	—	Reserved	The read value is 0. The write value should be 0.	R/W

The IPGR register sets the interpacket gap (IPG) value.

Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled) or while the ECMR.RE bit is 1 (receive function is enabled).

Refer to section 36.3.6, Adjusting Transmission Efficiency by Changing the IPG for details on the IPG.

36.2.9 Automatic PAUSE Frame Register (APR)

Address(es): ETHERC0.APR 000C 0154h, ETHERC1.APR 000C 0354h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	AP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	AP[15:0]	Automatic PAUSE Time Setting	These bits set the value of the pause_time parameter for a PAUSE frame that is automatically transmitted. Transmission is not performed until the set value multiplied by 512 bit time has elapsed.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The APR register sets the PAUSE time of the PAUSE frame that is automatically transmitted. The value set in the APR register is used for the pause_time parameter of the PAUSE frame.

Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled) or while the ECMR.RE bit is 1 (receive function is enabled).

36.2.10 Manual PAUSE Frame Register (MPR)

Address(es): ETHERC0.MPR 000C 0158h, ETHERC1.MPR 000C 0358h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	MP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	MP[15:0]	Manual PAUSE Time Setting	These bits set the value of the pause_time parameter for a PAUSE frame that is manually transmitted. Transmission is not performed until the set value multiplied by 512 bit time has elapsed. The read value is undefined.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The MPR register sets the PAUSE time of the PAUSE frame that is manually transmitted. The value set in the MPR register is used for the pause_time parameter of the PAUSE frame.
When a value is set to this register, a PAUSE frame is transmitted. Rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled).

36.2.11 Received PAUSE Frame Counter (RFCF)

Address(es): ETHERC0.RFCF 000C 0160h, ETHERC1.RFCF 000C 0360h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	RPAUSE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	RPAUSE[7:0]	Received PAUSE Frame Count	Number of received PAUSE frames	R
b31 to b8	—	Reserved	The read value is 0.	R

The RFCF register is a counter indicating the number of received PAUSE frames. The counter is reset after this register is read.

36.2.12 PAUSE Frame Retransmit Count Setting Register (TPAUSER)

Address(es): ETHERC0.TPAUSER 000C 0164h, ETHERC1.TPAUSER 000C 0364h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	TPAUSE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	TPAUSE[15:0]	Automatic PAUSE Frame Retransmit Setting	0000h: Number of retransmissions is unlimited 0001h: Maximum number of retransmissions is 1 : : FFFFh: Maximum number of retransmissions is 65,535	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The TPAUSER register selects the maximum number of times a PAUSE frame is automatically transmitted.
Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled).

36.2.13 PAUSE Frame Retransmit Counter (TPAUSECR)

Address(es): ETHERC0.TPAUSECR 000C 0168h, ETHERC1.TPAUSECR 000C 0368h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	TXP[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	TXP[7:0]	PAUSE Frame Retransmit Count	Number of times a PAUSE frame was retransmitted	R
b31 to b8	—	Reserved	The read value is 0.	R

The TPAUSECR register is a counter indicating the number of times a PAUSE frame was automatically retransmitted.
The counter is reset after this register is read.

36.2.14 Broadcast Frame Receive Count Setting Register (BCFRR)

Address(es): ETHERC0.BCFRR 000C 016Ch, ETHERC1.BCFRR 000C 036Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	BCF[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	BCF[15:0]	Broadcast Frame Continuous Receive Count Setting	0000h: Number of receptions is unlimited. 0001h: Receive 1 frame. : : FFFFh: Receive 65,535 frames.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The BCFRR register sets the number of times broadcast frames can be received continuously. When the number of received frames exceeds the BCF[15:0] bit value, the ECSR.BFR flag becomes 1 and the excess broadcast frames are discarded. The internal counter that counts the number of continuously received broadcast frames is reset when receiving any other frame than broadcast frame.

Do not rewrite this register while the ECMR.RE bit is 1 (receive function is enabled).

36.2.15 MAC Address Upper Bit Register (MAHR)

Address(es): ETHERC0.MAHR 000C 01C0h, ETHERC1.MAHR 000C 03C0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The MAHR register sets the upper 32 bits (b47 to b16) of the 48-bit MAC address. For example, if the MAC address is 01-23-45-67-89-AB, set the register to 0123 4567h.

Set the MAHR register during initialization after a reset. Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled) or while the ECMR.RE bit is 1 (receive function is enabled). When rewriting this register, set the EDMACn.EDMR.SWR bit to 1 to reset the EDMAC and ETHERC and then set this register again.

36.2.16 MAC Address Lower Bit Register (MALR)

Address(es): ETHERC0.MALR 000C 01C8h, ETHERC1.MALR 000C 03C8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	MA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	MA[15:0]	MAC Address Low	These bits set the lower 16 bits of the MAC address.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The MALR register sets the lower 16 bits of the 48-bit MAC address. For example, if the MAC address is 01-23-45-67-89-AB, set the register to 0000 89ABh.
Set the MALR register during initialization after a reset. Do not rewrite this register while the ECMR.TE bit is 1 (transmit function is enabled) or while the ECMR.RE bit is 1 (receive function is enabled). When rewriting this register, set the EDMACn.EDMR.SWR bit to 1 to reset the EDMAC and ETHERC and then set this register again.

36.2.17 Transmit Retry Over Counter Register (TROCR)

Address(es): ETHERC0.TROCR 000C 01D0h, ETHERC1.TROCR 000C 03D0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The TROCR register is a counter indicating the number of frames that fail to be retransmitted.
The TROCR register is incremented by 1 when a frame fails to be retransmitted 15 times. The counter stops when the TROCR register value becomes FFFF FFFFh. The counter value becomes 0 by writing any value to the TROCR register.

36.2.18 Late Collision Detect Counter Register (CDCR)

Address(es): ETHERC0.CDCR 000C 01D4h, ETHERC1.CDCR 000C 03D4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The CDCR register is a counter indicating the number of late collisions that have been detected after transmission starts. When the CDCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the CDCR register.

36.2.19 Lost Carrier Counter Register (LCCR)

Address(es): ETHERC0.LCCR 000C 01D8h, ETHERC1.LCCR 000C 03D8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The LCCR register is a counter indicating the number of times a loss of carrier is detected during frame transmission. When the LCCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the LCCR register.

36.2.20 Carrier Not Detect Counter Register (CNDCR)

Address(es): ETHERC0.CNDCR 000C 01DCh, ETHERC1.CNDCR 000C 03DCh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The CNDCR register is a counter indicating the number of times a carrier is not detected during preamble transmission. When the CNDCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the CNDCR register.

36.2.21 CRC Error Frame Receive Counter Register (CEFCR)

Address(es): ETHERC0.CEFCR 000C 01E4h, ETHERC1.CEFCR 000C 03E4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The CEFCR register is a counter indicating the number of received frames where a CRC error has been detected. When the CEFCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the CEFCR register.

36.2.22 Frame Receive Error Counter Register (FRECR)

Address(es): ETHERC0.FRECR 000C 01E8h, ETHERC1.FRECR 000C 03E8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The FRECR register is a counter indicating the number of times a frame receive error has occurred. The PHY-LSI notifies the ETHERC of the frame receive error using the ETn_RX_ER pin.

The FRECR register is incremented each time the ETn_RX_ER pin becomes high. When the FRECR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the FRECR register. When a frame receive error occurs while the EPTPC.SYBYPSR.BYPASSn bit is 0, the EPTPCn.SYSR.INFABT flag may become 1. In this case, reset the EPTPC, PTPEDMAC, and the corresponding channels ETHERC and EDMAC. Refer to section 36.5.4, Procedure to Reset Ethernet Controller for the procedure to reset these modules.

36.2.23 Too-Short Frame Receive Counter Register (TSFRCR)

Address(es): ETHERC0.TSFRCR 000C 01ECh, ETHERC1.TSFRCR 000C 03ECh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The TSFRCR register is a counter indicating the number of times a short frame that is shorter than 64 bytes has been received. When the TSFRCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the TSFRCR register.

36.2.24 Too-Long Frame Receive Counter Register (TLFRCR)

Address(es): ETHERC0.TLFRCR 000C 01F0h, ETHERC1.TLFRCR 000C 03F0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The TLFRCR register is a counter indicating the number of times a long frame that is longer than the RFLR register value has been received.
When the TLFRCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the TLFRCR register.
Note that the TLFRCR register is not incremented when a frame is received with the alignment error. In this case, the RFCR register is incremented.

36.2.25 Received Alignment Error Frame Counter Register (RFCR)

Address(es): ETHERC0.RFCR 000C 01F4h, ETHERC1.RFCR 000C 03F4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The RFCR register is a counter indicating the number of times a frame has been received with the alignment error (frame is not an integral number of octets).
When the RFCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the RFCR register.
When a frame is received with an alignment error while the EPTPC.SYBYPSR.BYPASSn bit is 0, the EPTPCn.SYSR.INFABT flag may become 1. In this case, reset the EPTPC, PTPEDMAC, and the corresponding channels ETHERC and EDMAC. Refer to section 36.5.4, Procedure to Reset Ethernet Controller for the procedure to reset these modules.

36.2.26 Multicast Address Frame Receive Counter Register (MAFCR)

Address(es): ETHERC0.MAFCR 000C 01F8h, ETHERC1.MAFCR 000C 03F8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The MAFCR register is a counter indicating the number of times a frame where the multicast address is set has been received.

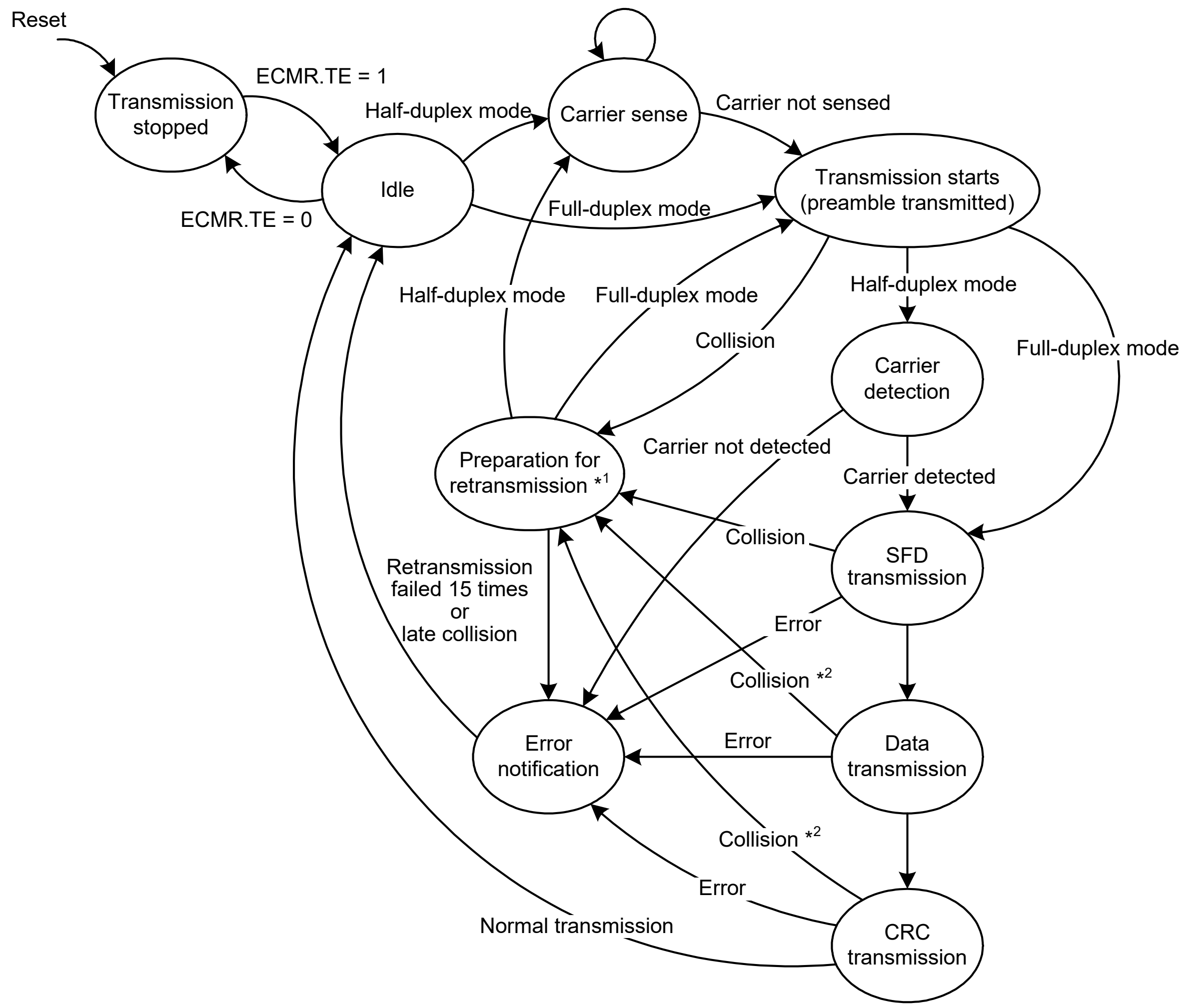
When the RFCR register value becomes FFFF FFFFh, the counter stops. The counter value becomes 0 by writing any value to the RFCR register.

36.3 Operation

This section is an overview of the ETHERC operations. The ETHERC supports flow control compliant with IEEE802.3x, and can transmit and receive PAUSE frames.

36.3.1 Transmission

The ETHERC transmitter assembles transmit data into a frame and outputs it to the MII/RMII when a transmit request is received from the EDMAC. The frame transmitted via the MII/RMII is transmitted on the line by the PHY-LSI. Figure 36.4 shows the state transitions of the ETHERC transmitter.



Note 1. Preparation for retransmission includes transmitting a jam signal and adjusting transmission time intervals using the backoff algorithm after a collision is detected.

Note 2. Preparation for retransmission is performed only when a collision is detected within 512 bit time after preamble transmission starts. When a collision is detected after 512 bit time, only a jam signal is transmitted and processing using the backoff algorithm is not performed.

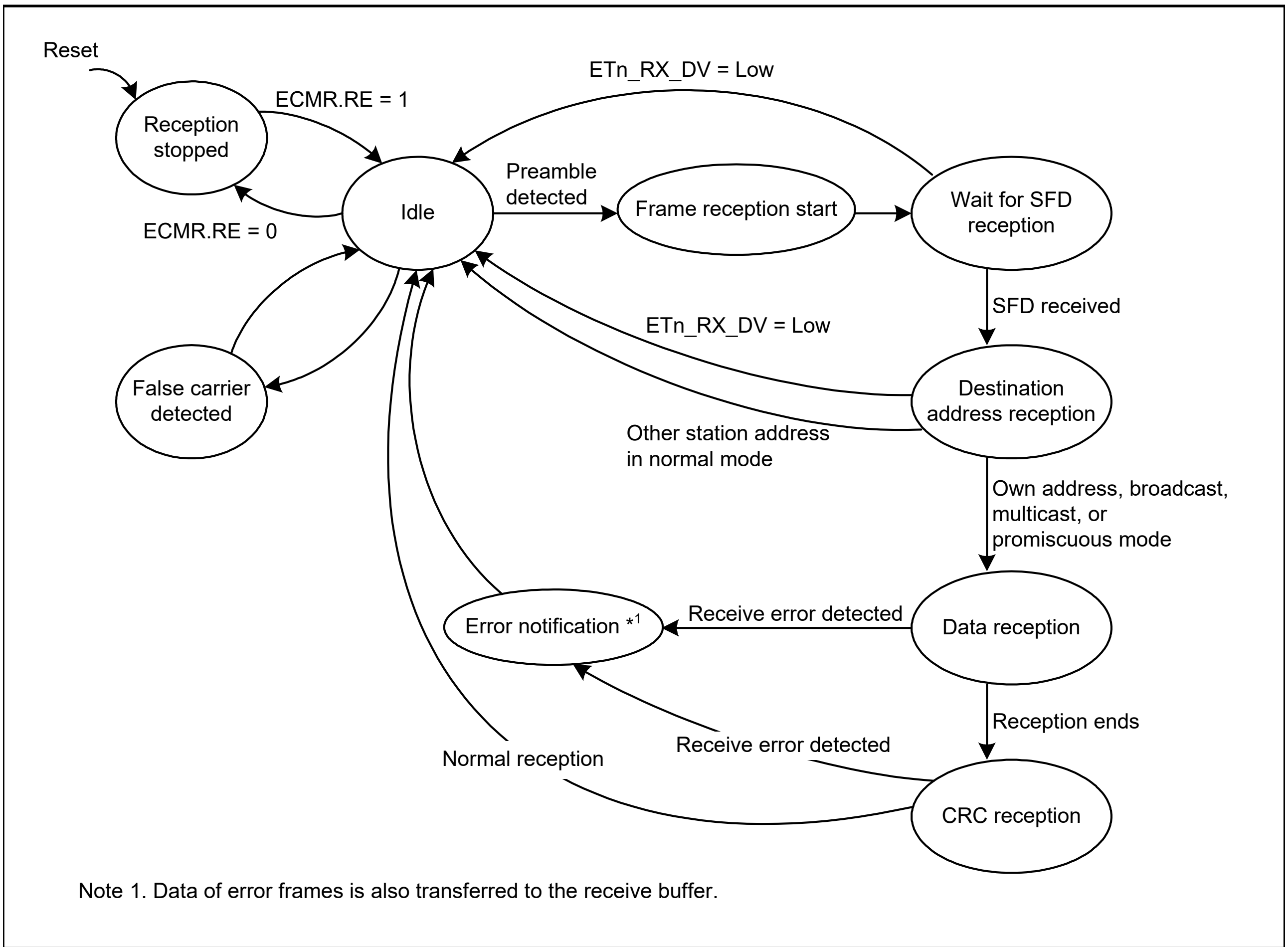
Figure 36.4 ETHERC Transmitter State Transitions

1. When setting the ECMR.TE bit to 1, the ETHERC enters the transmit idle state.
2. When a transmit request is received from the EDMAC, the ETHERC enters the carrier sense state. The ETHERC waits for the interpacket gap and then transmits a preamble to the MII/RMII. When full-duplex mode is selected, it is not required to sense a carrier, so the ETHERC transmits a preamble immediately after receiving a transmit request from the EDMAC.

3. The ETHERC transmits the Start Frame Delimiter (SFD), transmit data, and CRC sequentially. When the transmission is completed successfully, the ETHERC notifies the EDMAC of successful completion, and the EDMAC sets the EDMACn.EESR.TC flag to 1. When a late collision or loss of carrier is detected during data transmission, the ETHERC stops the transmission and notifies the EDMAC of the error.
4. After the time for the interpacket gap has elapsed, the ETHERC enters the idle state and continues transmission when transmit data remains.

36.3.2 Reception

The ETHERC receiver separates the frame input from the MII/RMII into the preamble, SFD, receive data, and CRC, and transmits only receive data (destination address, source address, type/length, data/LLC). Figure 36.5 shows the state transitions of the ETHERC receiver.



Note 1. Data of error frames is also transferred to the receive buffer.

Figure 36.5 ETHERC Receiver State Transitions

1. When setting the ECMR.RE bit to 1, the ETHERC enters the receive idle state.
2. When the SFD following the preamble of the receive packet is detected, the ETHERC starts reception. If the received SFD is invalid, the ETHERC discards the frame.
3. In normal mode, the ETHERC starts data reception when the destination address of the receive frame is the address of the MCU or the receive frame is broadcast frame or multicast frame. In promiscuous mode, the ETHERC starts data reception regardless of the receive frame type.
4. After receiving data from the MII/RMII, the ETHERC performs the CRC check. The ETHERC notifies the EDMAC of the CRC check result. After the received data is transferred to the receive buffer, the CRC check result is written back to the receive descriptor as a status. The result is also reflected in the EDMACn.EESR.CERF flag.
5. When the ECMR.RE bit is 1 after one frame has been received, the ETHERC prepares to receive the next frame.

36.3.3 Frame Timing

36.3.3.1 MII Frame Timing

Figure 36.6 to Figure 36.10 show the MII frame timing.

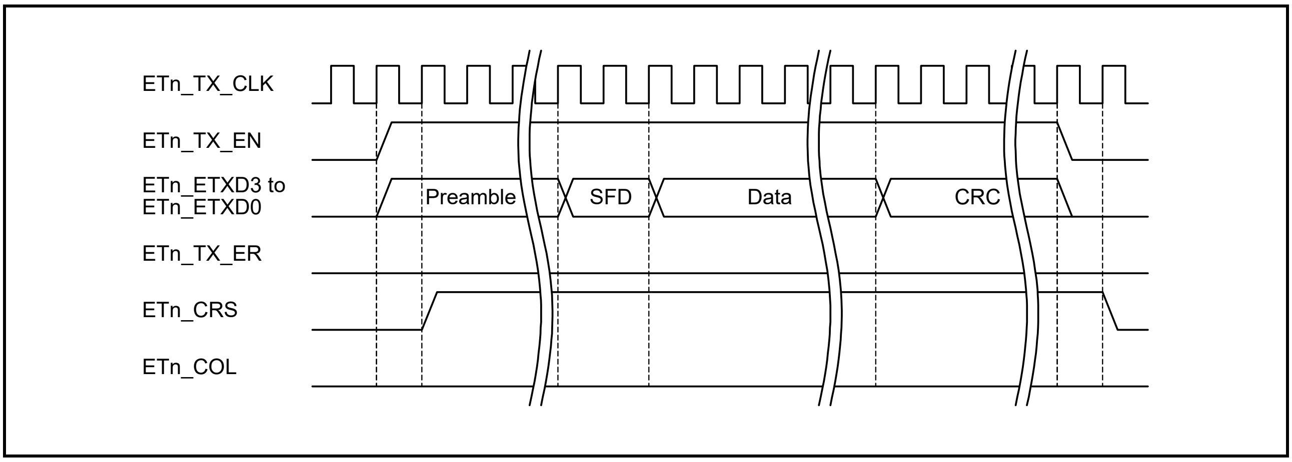


Figure 36.6 MII Frame Transmit Timing During Normal Transmission

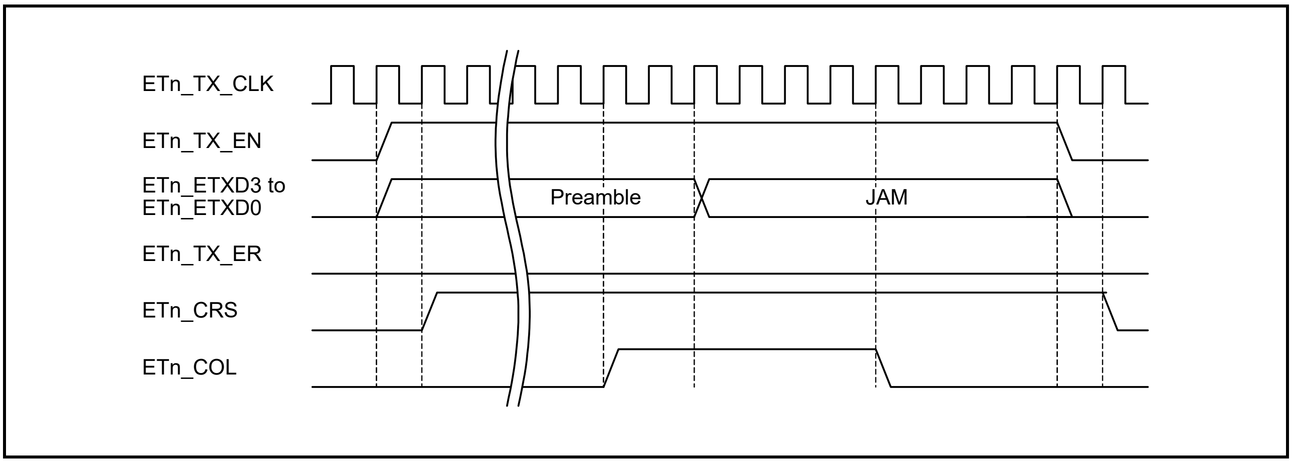


Figure 36.7 MII Frame Transmit Timing When Collision Occurs

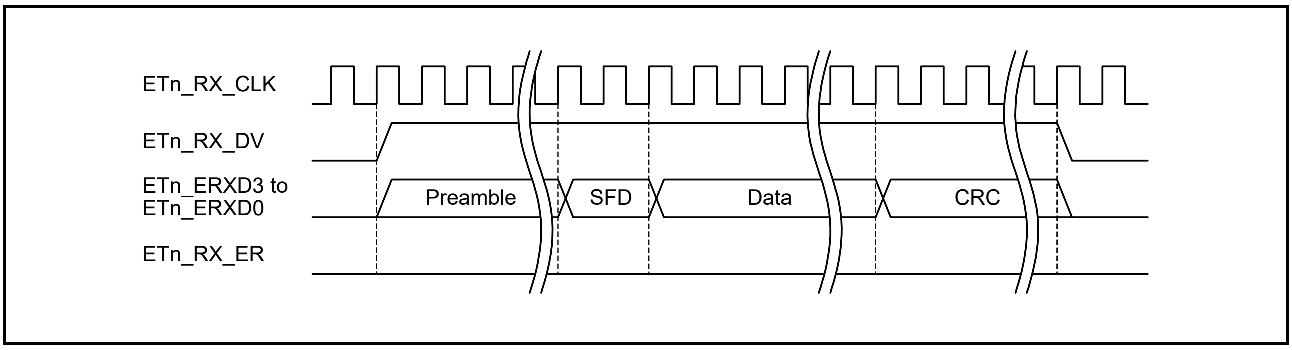


Figure 36.8 MII Frame Receive Timing During Normal Reception

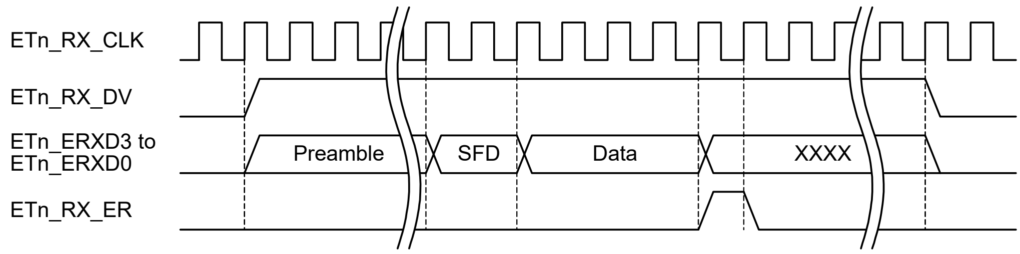


Figure 36.9 MII Frame Receive Timing for Receive Error Notification

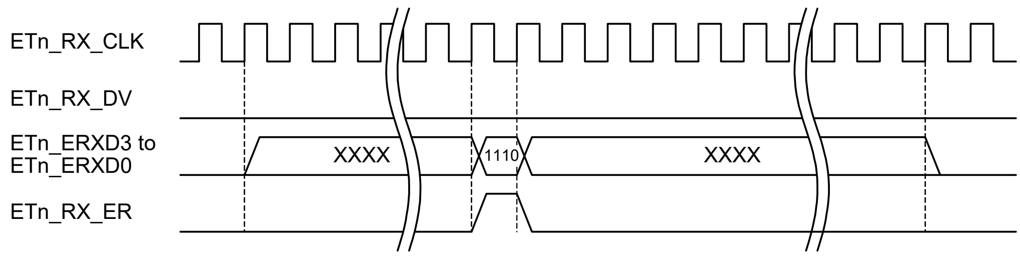


Figure 36.10 MII Fame Receive Timing for False Carrier Notification

36.3.3.2 RMII Frame Timing

The RMII frame timing is shown in Figure 36.11 to Figure 36.13.

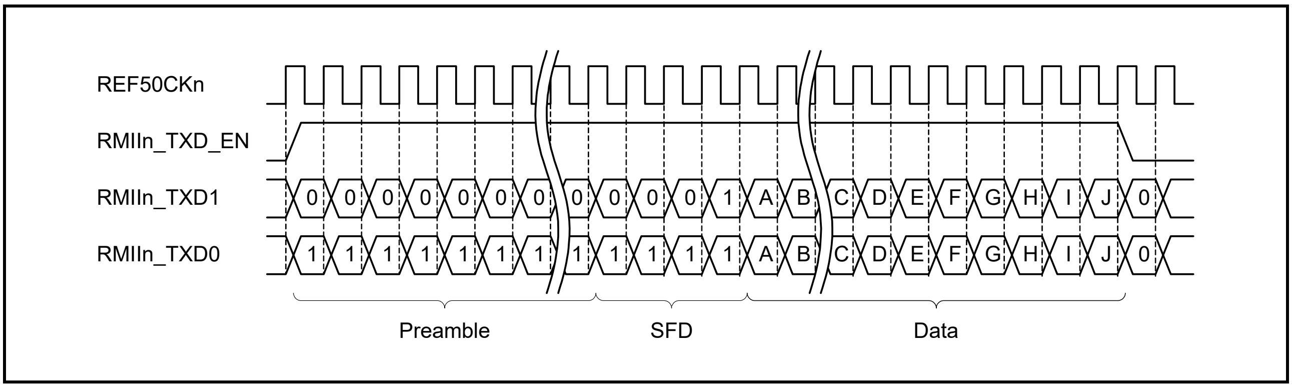


Figure 36.11 RMII Frame Transmit Timing During Normal Transmission

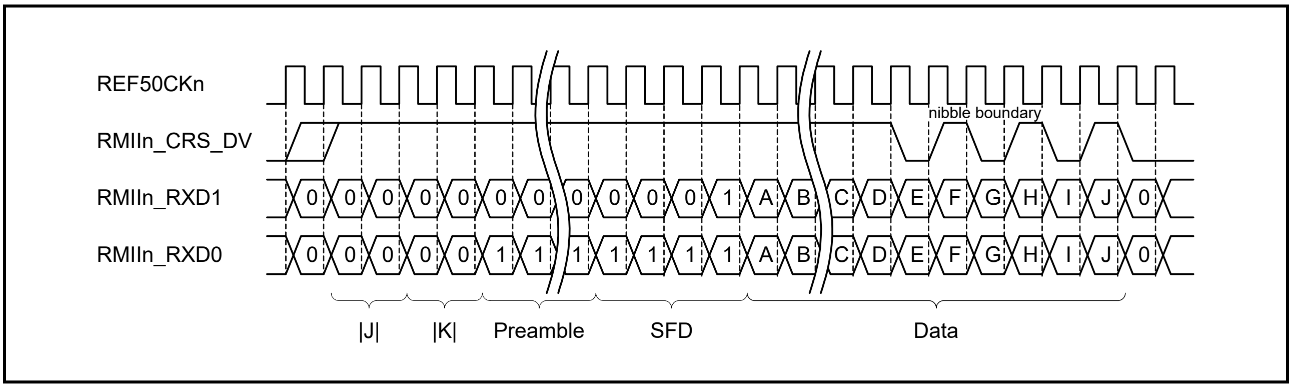


Figure 36.12 RMII Frame Receive Timing During Normal Reception

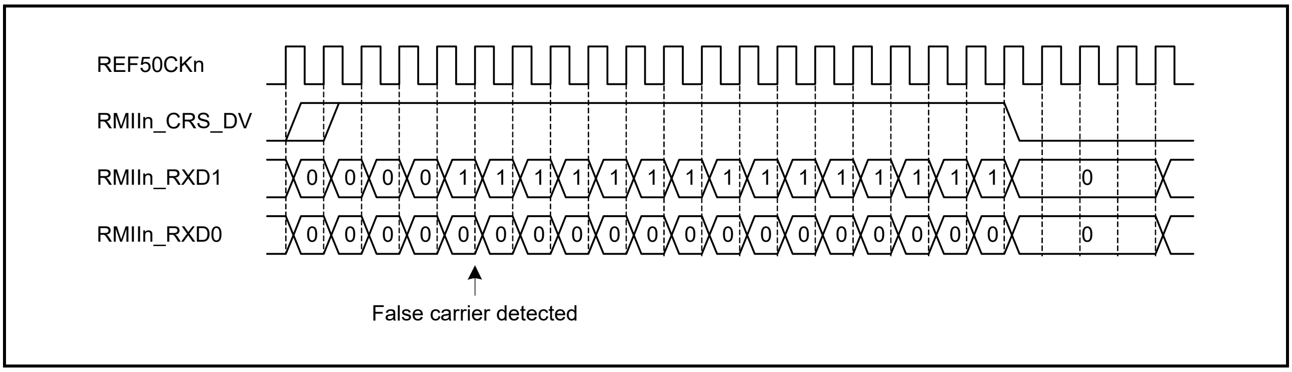


Figure 36.13 RMII Frame Receive Timing When False Carrier Is Detected

36.3.4 Accessing MII/RMII Registers

Use the PIR register to access the MII/RMII registers in the PHY-LSI. Serial data in the MII/RMII management frame format is transmitted and received via the ETn_MDC and ETn_MDIO pins controlled by software.

36.3.4.1 MII/RMII Management Frame Format

Table 36.3 lists the MII/RMII management frame format.

Table 36.3 MII/RMII Management Frame Format

Access Type	MII/RMII Management Frame							
Item	PRE	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
Number of bits	32	2	2	5	5	2	16	1
Read	1...1	01	10	00001	RRRRR	Z0	DDDDDDDDDDDDDDDD	Z
Write	1...1	01	01	00001	RRRRR	10	DDDDDDDDDDDDDDDD	Z

PRE (preamble):	Send 32 consecutive 1's.
ST (start of frame):	Send 01b.
OP (operation code):	Send 10b for read or 01b for write.
PHYAD (PHY address):	Up to 32 PHY-LSIs can be connected to one MAC. PHY-LSIs are selected with these 5 bits. When the PHY-LSI address is 1, send 00001b.
REGAD (register address):	One register is selected from up to 32 registers in the PHY-LSI. When the register address is 1, send 00001b.
TA (turnaround):	2-bit turnaround time to avoid contention between the register address and data during a read operation (a) Send 10b during a write operation. (b) Release the bus for 1 bit during a read operation (Z is output) (indicated as Z0 because 0 is output from the PHY-LSI on the next clock cycle).
DATA (data):	16-bit data. Sequentially send or receive starting from the MSB.
IDLE (IDLE condition):	Wait time to input the next MII/RMII management format (a) Release the bus during a write operation (Z is output). (b) No control is required since a bus has already been released during a read operation.

36.3.4.2 MII/RMII Register Access Procedure

Access to the MII/RMII registers includes writing data in 1-bit units, reading data in 1-bit units, and releasing the bus. Figure 36.14 to Figure 36.17 show examples of the MII/RMII register access timing. The access timing differ with the PHY-LSI type.

(1) Write to the PHY interface register
PIR.MMD bit ← 1 (write)
PIR.MDO bit ← write data
PIR.MDC bit ← 0

(2) Write to the PHY interface register
PIR.MMD bit ← 1 (write)
PIR.MDO bit ← write data
PIR.MDC bit ← 1

(3) Write to the PHY interface register
PIR.MMD bit ← 1 (write)
PIR.MDO bit ← write data
PIR.MDC bit ← 0

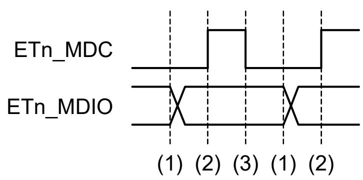


Figure 36.14 1-Bit Data Write Flow

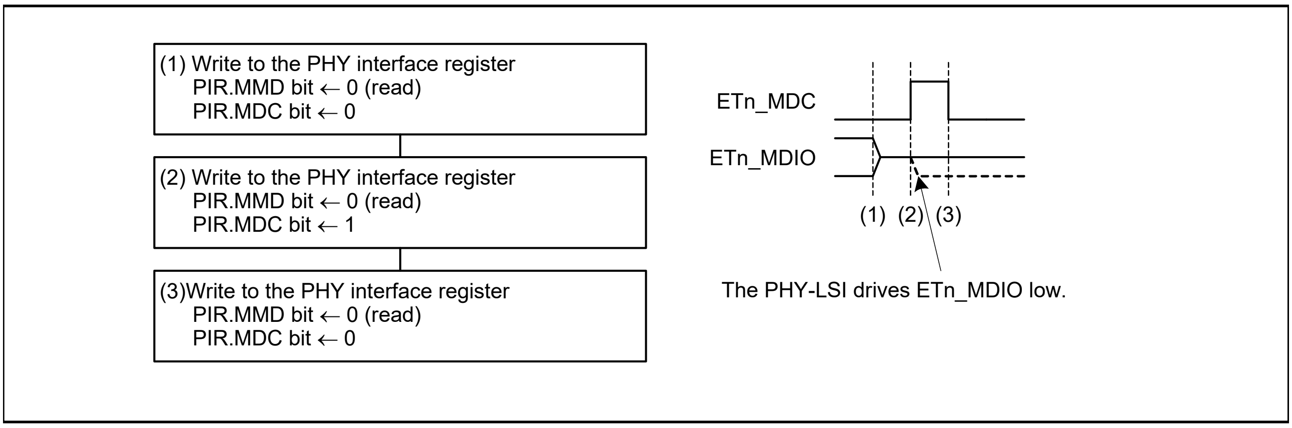


Figure 36.15 Bus Release Flow (TA in Read Operation in Table 36.3)

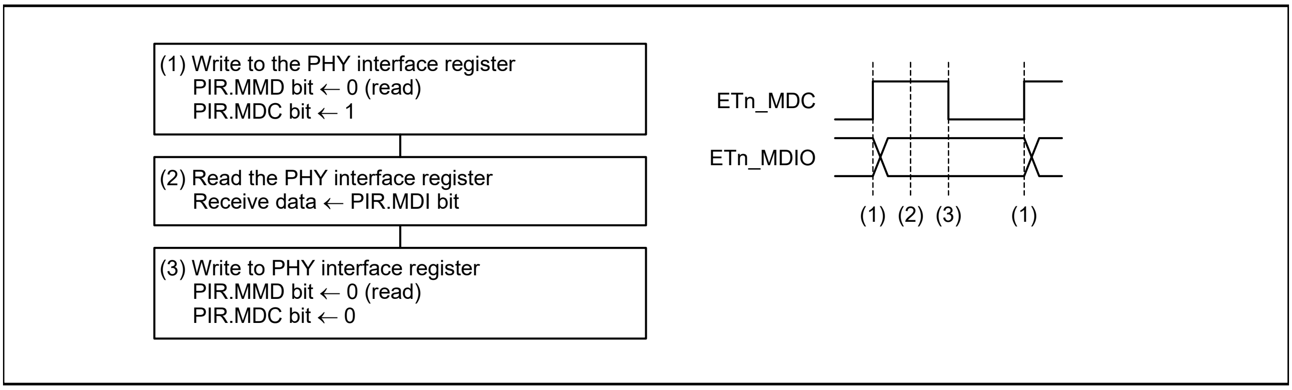


Figure 36.16 1-Bit Data Read Flow

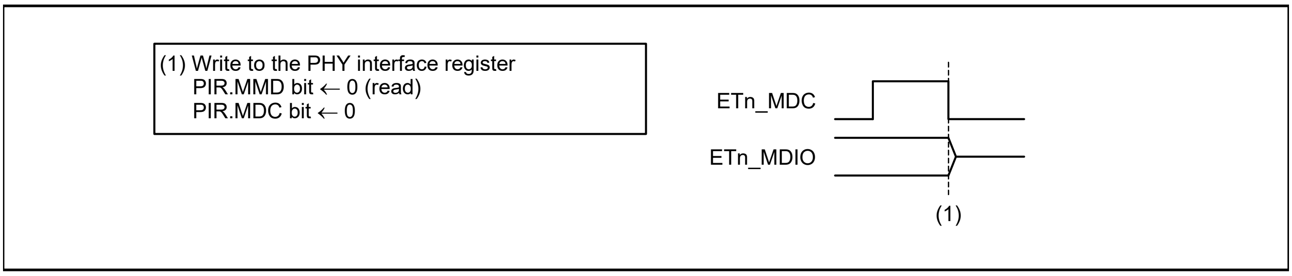


Figure 36.17 Bus Release Flow (IDLE in Write Operation in Table 36.3)

36.3.5 Magic Packet Detection

The ETHERC supports Wake-On-LAN (WOL). WOL is a function to detect a Magic Packet transmitted from a host device or other device and exit a low power consumption state such as sleep mode. When the ETHERC detects a Magic Packet, high is output from the ETn_WOL pin. Write 1 to the EDMACn.EDMR.SWR bit to set the ETn_WOL pin to low.
Since a Magic Packet is transmitted in broadcast mode, the Magic Packet is received regardless of the destination MAC address selected in the format. The ETHERC outputs high from the ETn_WOL pin only when the destination MAC address matches its own MAC address. Refer to the technical documentation provided by Advanced Micro Devices, Inc. for details on the Magic Packet. The following describes an example of the procedure to use WOL in the MCU.

1. Set the ICU to disable the EINTn interrupt request.
2. Set the ECMR.MPDE bit to 1 to enable Magic Packet detection. Set the ECMR.RE bit to 1 to enable reception.
3. Set the ECSIPR.MPDIP bit to 1 to enable notification of the Magic Packet detect interrupt.
4. Set the EDMACn.EESIPR.ECIIP bit to 1 to enable the ETHERC status register source interrupt.
5. Set the ICU to enable the EINTn interrupt request.
6. Change the CPU operating mode to sleep mode or place unused peripherals in the module stop state as needed.
7. When a Magic Packet is detected, an interrupt request is sent to the CPU. High is output from the ETn_WOL pin to notify peripheral devices that the Magic Packet has been detected.

36.3.5.1 Notes on Magic Packet Detection

The ETHERC receives packets, including broadcast packets, even when waiting to receive a Magic Packet. Therefore, receive data may have been stored in the receive FIFO of the EDMAC when a Magic Packet is detected. Also, flags in registers ECSR and EDMACn.EESR may have been changed. When returning to normal operation by detecting a Magic Packet, set the EDMACn.EDMR.SWR bit to 1 to reset the ETHERC and EDMAC.

36.3.6 Adjusting Transmission Efficiency by Changing the IPG

The IPG is a non-transmit period between transmit frames. The ETHERC can change the value of the IPG. Transmission efficiency can be increased or decreased by setting the IPGR register. The typical value of the IPG is specified by the IEEE802.3 standard. When changing the setting, confirm that all devices operate normally in the same network.

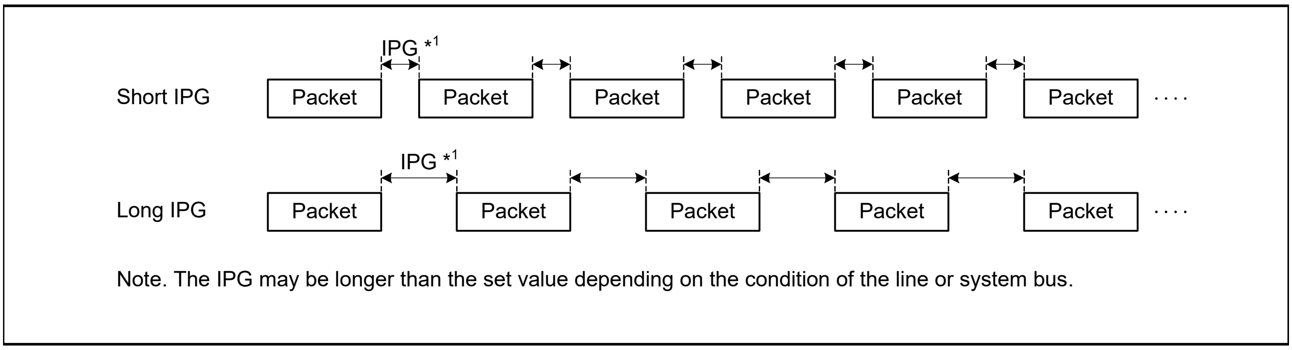


Figure 36.18 Differences in Transmission Efficiency Based on Changes in the IPG

36.3.7 Flow Control

The ETHERC can perform flow control compliant with IEEE802.3x in full-duplex mode, and the receiver and transmitter can be set individually. PAUSE frames can be transmitted automatically or manually.

36.3.7.1 Automatic PAUSE Frame Transmission

When the ECMR.TXF bit is set to 1, automatic PAUSE frame transmission is enabled. A PAUSE frame is automatically transmitted by a PAUSE frame transmit request from the EDMAC. The APR.AP[15:0] bit value is used for the pause_time parameter of the PAUSE frame.
After a PAUSE frame is transmitted, if the EDMAC is still requesting PAUSE frame transmission when the PAUSE time elapses, a PAUSE frame is transmitted again. The maximum number of PAUSE frame retransmissions can be set in the TPAUSER.TPAUSE[15:0] bits. If the maximum number of retransmissions is reached, subsequent PAUSE frames are not transmitted.
Figure 36.19 shows the procedure to set automatic PAUSE frame transmission.

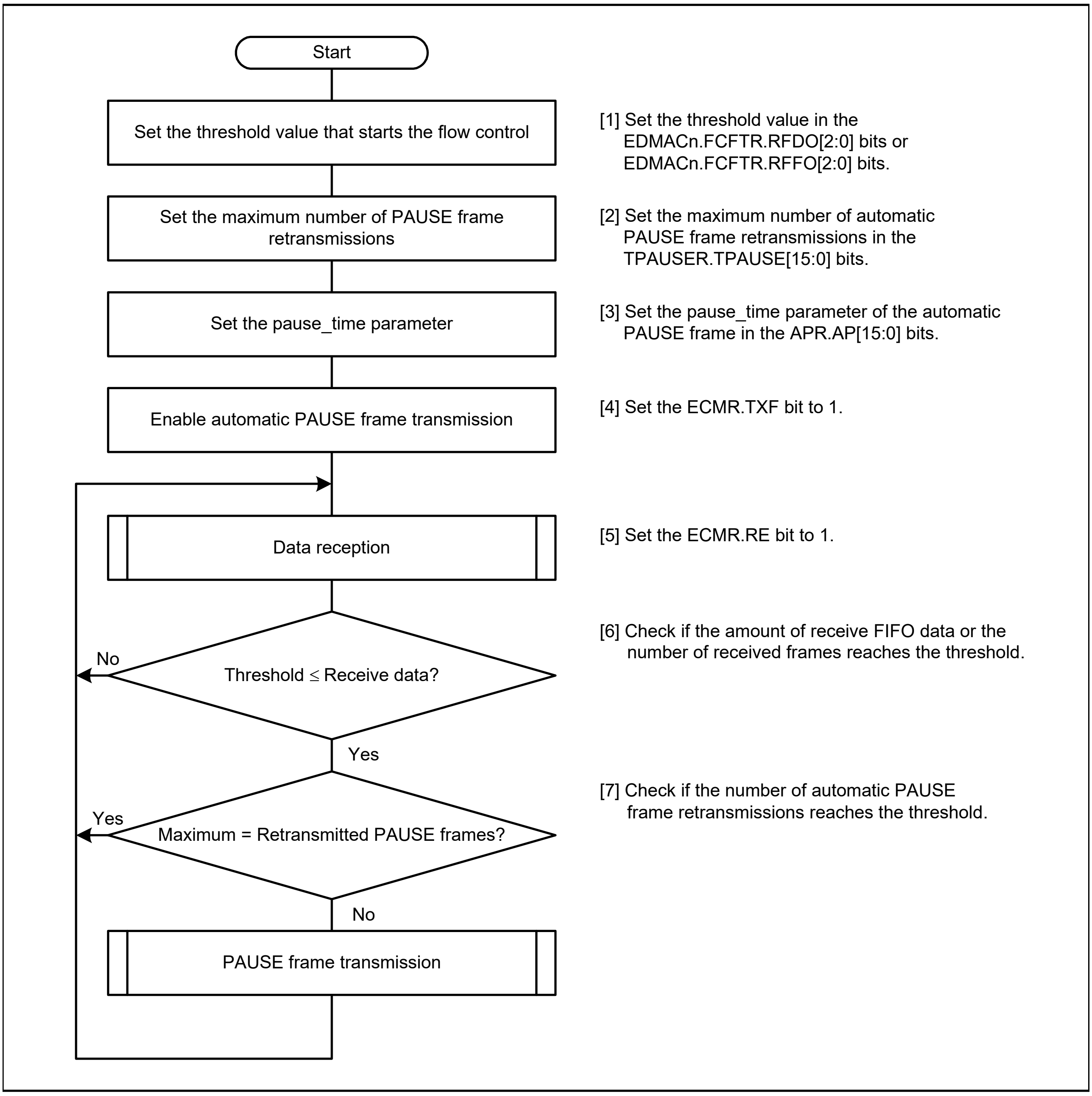


Figure 36.19 Example of Procedure to Set Automatic PAUSE Frame Transmission

36.3.7.2 Manual PAUSE Frame Transmission

A PAUSE frame can be manually transmitted at any time. When writing the pause_time parameter of the PAUSE frame to the MPR.MP[15:0] bits by software, the ETHERC transmits a PAUSE frame once. When transmitting a PAUSE frame more than once, write to the MPR.MP[15:0] bits for each transmission.

36.3.7.3 PAUSE Frame Reception

When setting the ECMR.RXF bit to 1, PAUSE frame detection is enabled. After a PAUSE frame is received, the ETHERC completes transmitting the current frame and waits for the PAUSE time of the received PAUSE frame to elapse before the next frame can be transmitted. Also, the ETHERC increments the RFCF.RPAUSE[7:0] bit value.
However, while waiting for the PAUSE time to elapse, if a PAUSE frame that contains a pause_time parameter of 0 is received and the ECMR.ZPF bit is 1, the ETHERC becomes ready to transmit.

36.4 Interrupts

When a flag in the ECSR register becomes 1 and the corresponding bit in the ECSIPR register is 1, the ETHERC notifies the EDMAC of the status as an interrupt source. After receiving the notification, the EDMAC sets the EDMACn.EESR.ECI flag to 1. When the EDMACn.EESIPR.ECIIP bit is 1, the EDMAC sends an EINTn interrupt request to the CPU. Refer to section 38, DMA Controller for the Ethernet Controller (EDMACa) for details.

36.5 Usage Notes

36.5.1 Conditions for the LCHNG Flag to Become 1

The ECSR.LCHNG flag may become 1 even when the input level of the ETn_LINKSTA pin remains the same. In this case, high is input to the ETn_LINKSTA pin when setting the MPC.PmnPFS register to assign the ETn_LINKSTA signal to a port or when releasing the ETHERC and EDMAC software reset using the EDMACn.EDMR.SWR bit. The ECSR.LCHNG flag becomes 1 because the ETn_LINKSTA signal in the ETHERC is fixed low regardless of the input level to the external pin while the MPC does not assign the ETn_LINKSTA signal or during the ETHERC and EDMAC software reset.
To avoid wrongly generating a link signal change interrupt, clear the ECSR.LCHNG flag and then set the ECSIPR.LCHNGIP bit to 1.

36.5.2 Input to the RMIIn_RX_ER Pin While the RMII is Selected

When the width of a reception error signal received from the PHY-LSI is only one cycle of the REF50CKn clock (50 MHz) while the RMII is selected, the signal is not recognized as an error signal.

36.5.3 Handling Errors in Control Information

When a frame with an alignment error is received or a frame receive error occurs while the EPTPC.SYBYPSR.BYPASSn bit is 0, the EPTPCn.SYSR.INFABT flag may become 1. In this case, reset the EPTPC, PTPEDMAC, and the corresponding channels ETHERC and EDMAC. Refer to section 36.5.4, Procedure to Reset Ethernet Controller for the procedure to reset these modules.

36.5.4 Procedure to Reset Ethernet Controller

Follow the procedure below to reset the ETHERC, EPTPC, and EDMAC. In this case, data in the address range from 0000 0000h to 0000 001Fh may be destroyed.

(a) When the EPTPC is not in use

(1) Write 0000 0001h (reset ETHERCn and EDMACn) to the EDMACn.EDMR register of the corresponding channels.
(2) Wait for 64 cycles of the PCLKA until initialization is completed.

(b) When the EPTPC is in use

(1) Write 0000 0001h (reset EPTPC) to the EPTPC.PTRSTR register.
(2) Write 0000 0001h (reset PTPEDMAC) to the PTPEDMAC.EDMR register.
(3) Write 0000 0001h (reset ETHERCn and EDMACn) to the EDMACn.EDMR register of the corresponding channels.
(4) Wait for 64 cycles of the PCLKA until initialization is completed.
(5) Write 0000 0000h (release reset of EPTPC) to the EPTPC.PTRSTR register.
(6) Wait for 256 cycles of the PCLKA until the EPTPC is released from the reset.